SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

NSS Management Portal

Gautam Buddha University, Greater Noida, Uttar Pradesh

Document Version: 1.0
Status: Draft for Development and Approval
System Type: Internal Web-Based Management Portal
Intended Users: NSS Programme Coordinator, Programme Officers, University Administration, Finance/Accounts Staff, Inventory Manager and NSS Volunteers
Initial Scale: 6 NSS Units, up to 100 volunteers per unit (600 volunteers total)

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional, technical, operational and security requirements for developing an integrated National Service Scheme (NSS) Management Portal for Gautam Buddha University (GBU).

The portal shall digitize the complete NSS volunteer lifecycle, including registration, unit allocation, activity management, attendance, service-hour tracking, special camps, performance evaluation, certificate management, financial management, inventory management and institutional reporting.

The document shall serve as the primary reference for the software developer, UI/UX designer, database developer, tester and university's authorized project coordinator.

1.2 Project Objectives

The system shall:

1. Maintain a centralized and reliable database of all NSS volunteers.
2. Manage six NSS units and their respective Programme Officers.
3. Record activities, attendance and verified service hours throughout the volunteer's NSS tenure.
4. Track completion of prescribed service-hour requirements and special-camp requirements.
5. Identify active, consistent, high-performing and inactive volunteers.
6. Manage events, online activities, quizzes, leaderboards and student engagement.
7. Generate, approve, issue, record and verify certificates through authorized workflows.
8. Maintain event-wise and unit-wise budgets, financial sanctions, expenditure, bills and vouchers.
9. Maintain inventory, stock movement, distribution and asset records.

10. Generate reports for NSS administration, annual reporting, university review, accreditation and authorized government reporting.
11. Maintain accountability through role-based access, audit trails and controlled approvals.

1.3 Scope

The project includes an internal portal with authenticated access, an administrative dashboard, a student/volunteer dashboard and role-specific operational interfaces.

Out of scope for the initial release:

- A separate public-facing NSS website.
- Automatic issuance of government certificates without authorized approval.
- Replacement of the university's official accounting, payment or government financial systems.
- Direct Instagram publishing without an approved integration and required permissions.

News, photos and social media records may be maintained internally for documentation and reporting.

1.4 Initial Capacity

- NSS Units: 6
- Initial capacity per unit: 100 volunteers
- Initial total capacity: 600 volunteers
- Service period: Two years, subject to applicable NSS rules
- Service hours: Configurable annual and cumulative requirements
- Special Camp: Configurable duration, including the prescribed seven-day camp where applicable

Unit capacity, service-hour requirements, activity categories and certificate eligibility criteria shall be configurable by authorized administrators rather than hard-coded.

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2. User Roles and Access Control

The system shall implement Role-Based Access Control (RBAC).

2.1 Super Administrator / IT Administrator

- Manage user accounts, roles, permissions and system configuration.
- Create and deactivate NSS units.
- Configure academic sessions and reporting periods.
- Manage authentication and system-level settings.

- View audit logs and system activity.
- Perform approved data imports and exports.

The IT Administrator shall not automatically receive authority to approve financial transactions or issue certificates unless those permissions are explicitly assigned.

2.2 NSS Programme Coordinator

- View and manage all NSS units.
- Monitor volunteer enrollment and participation.
- Review activity and attendance summaries.
- Monitor service-hour completion and special-camp status.
- Review performance dashboards and unit rankings.
- Approve designated operational workflows.
- Review and recommend certificate cases according to the approved delegation of authority.
- Generate consolidated reports.

2.3 NSS Programme Officer

- Manage volunteers assigned to the officer's unit.
- Create activities and event registrations.
- Record and verify attendance.
- Review and approve service-hour entries within delegated authority.
- Upload activity reports, photographs and supporting documents.
- Monitor volunteer performance.
- Recommend eligible volunteers for certification.
- Submit event budgets and expenditure records where authorized.

2.4 Finance / Accounts Officer

- Review budget requests.
- Record or verify sanctioned and released amounts.
- Review bills, vouchers and expenditure documents.
- Record payment and settlement status.
- Review utilization and outstanding balances.
- Generate financial statements and expenditure reports.
- Maintain financial approval records according to delegated authority.

2.5 Inventory Manager

- Maintain item and asset registers.
- Record purchases, receipts, issues, returns and stock adjustments.
- Assign inventory to events or units.
- Record damaged, lost or disposed items.
- Generate stock and asset reports.

2.6 Certificate Approving Authority

- Review eligibility and supporting records.
- Approve or reject certificate requests.
- Record approval remarks and date.
- Authorize certificate issuance.
- Revoke or supersede a certificate only through an authorized, audited process.

The final approving authority must be designated by the university in accordance with applicable rules.

2.7 Media / Content Manager

- Maintain internal news, photographs, videos and activity documentation.
- Record social media URLs and engagement metrics.
- Submit content for approval where required.

2.8 NSS Volunteer

- View and update permitted profile details.
- View assigned unit and service history.
- Register for eligible activities.
- View attendance, verified hours and pending corrections.
- Participate in online quizzes and training.
- View personal performance and certificate eligibility status.
- Download authorized certificates.

Volunteers shall not be permitted to modify approved attendance, service hours, financial records or certificates.

2.9 University Management / Read-Only Viewer

- View authorized dashboards and reports.
- Access aggregated statistics and approved documents.
- No modification rights unless separately assigned.

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3. Functional Requirements

Requirement IDs shall be retained during development and testing.

3.1 Module M01 — Authentication and User Management

FR-AUTH-01: The system shall provide secure login and logout.

FR-AUTH-02: The system shall support role-based permissions.

FR-AUTH-03: Each user shall have a unique account identifier.

FR-AUTH-04: The system shall support account activation, suspension and deactivation.

FR-AUTH-05: The system shall provide password reset and account recovery through approved mechanisms.

FR-AUTH-06: Unauthorized users shall not access protected pages or APIs.

FR-AUTH-07: Sensitive actions shall require re-authentication or additional verification where appropriate.

FR-AUTH-08: The system shall record successful and failed login attempts, relevant security events and account changes.

FR-AUTH-09: Users may have multiple authorized roles where required.

3.2 Module M02 — Volunteer Registration and Management

FR-VOL-01: The system shall provide an online volunteer registration form.

FR-VOL-02: The registration form shall capture:

- Full name and photograph.
- University enrollment/roll number.
- School, department, course and academic year.
- Mobile number and email.
- NSS joining date and academic session.
- Assigned NSS unit.
- Emergency contact where institutionally required.
- Consent and declaration fields where applicable.

FR-VOL-03: The system shall generate a unique NSS Volunteer ID.

FR-VOL-04: The system shall prevent duplicate enrollment using configured unique identifiers.

FR-VOL-05: Registration shall support Pending, Approved, Rejected and Waitlisted states.

FR-VOL-06: Unit allocation shall respect the configured capacity.

FR-VOL-07: Authorized users shall be able to import volunteer records through validated CSV/Excel templates.

FR-VOL-08: The system shall maintain volunteer status: Active, Inactive, Completed, Withdrawn or other administrator-defined statuses.

FR-VOL-09: The system shall preserve historical unit allocation and academic-session records.

FR-VOL-10: Authorized users shall be able to search and filter volunteers by unit, school, academic year, status and service-hour progress.

3.3 Module M03 — NSS Unit Management

FR-UNIT-01: The system shall support six initial NSS units.

FR-UNIT-02: Each unit shall have a unique unit code, name, capacity, assigned Programme Officer and status.

FR-UNIT-03: Authorized administrators shall be able to change unit capacity.

FR-UNIT-04: The system shall show allocated, available and waitlisted volunteer counts.

FR-UNIT-05: The system shall provide unit-wise activity, attendance, hours, camp participation and financial summaries.

FR-UNIT-06: Unit-level users shall only access records within their authorized scope.

3.4 Module M04 — Activity and Event Management

FR-EVT-01: Authorized users shall be able to create, edit, publish, cancel and close events.

FR-EVT-02: Each event shall have a unique Event ID.

FR-EVT-03: Event categories shall include:

- Campus Activity.
- Field Activity.
- Adopted Village Activity.
- Slum/Community Outreach.
- School Outreach.
- Special Camp.
- Online Activity.
- Training and Workshop.
- Awareness Campaign.
- University, State or National Level Event.
- Other administrator-defined categories.

FR-EVT-04: Each event shall capture:

- Title and description.
- Category and objectives.

- Date, start time and end time.
- Venue or online meeting details.
- Organizing unit(s).
- Responsible coordinator.
- Target and maximum participants.
- Assigned service hours and calculation method.
- Registration deadline.
- Attendance method.
- Budget and financial references.
- Required documents.
- Event status.

FR-EVT-05: Events shall support participation by multiple NSS units.

FR-EVT-06: The system shall support registration, cancellation and waiting lists.

FR-EVT-07: The system shall store event photographs, reports, attendance sheets and supporting evidence.

FR-EVT-08: The system shall maintain an event lifecycle: Draft, Submitted, Approved, Published, Ongoing, Completed, Cancelled or Archived.

FR-EVT-09: Changes to an approved event's dates, hours or financial details shall be recorded in the audit log.

3.5 Module M05 — Attendance and Service-Hour Ledger

FR-HRS-01: Attendance shall be recorded against a specific event and volunteer.

FR-HRS-02: Supported attendance methods shall include:

- Manual entry by authorized Programme Officers.
- QR-based check-in/check-out.
- Validated bulk attendance import.
- Approved online participation evidence.

FR-HRS-03: Attendance states shall include Present, Absent, Partial Attendance, Pending Verification and Approved.

FR-HRS-04: The system shall distinguish registered participants from verified attendees.

FR-HRS-05: Service-hour calculations shall consider the approved activity rules and verified participation.

FR-HRS-06: The system shall maintain a separate ledger entry for each approved volunteer-event contribution.

FR-HRS-07: Each ledger entry shall record:

- Volunteer ID.
- Event ID.
- Attendance reference.
- Proposed and approved hours.
- Approval status.
- Approving user and timestamp.
- Supporting evidence.
- Remarks and adjustment history.

FR-HRS-08: Duplicate hour credits for the same volunteer and event shall be prevented unless an authorized, documented exception is approved.

FR-HRS-09: The system shall calculate annual and cumulative hours for each volunteer.

FR-HRS-10: The system shall display required hours, approved hours, pending hours and remaining hours separately.

FR-HRS-11: Corrections to approved hours shall require authorized approval and an audit trail.

FR-HRS-12: The system shall preserve service-hour records across academic sessions.

FR-HRS-13: Service-hour targets and special-camp requirements shall be configurable according to the applicable NSS rules.

FR-HRS-14: Online quizzes or training shall not automatically count as NSS service hours unless the activity has been approved for such credit.

3.6 Module M06 — Special Camp Management

FR-CAMP-01: Authorized users shall be able to create a special-camp record.

FR-CAMP-02: The camp record shall include location, dates, duration, adopted village/community, participating units and responsible officers.

FR-CAMP-03: The system shall support day-wise schedules and activities.

FR-CAMP-04: Day-wise attendance shall be maintained for every participant.

FR-CAMP-05: The system shall record day-wise activities, service hours, photographs and reports.

FR-CAMP-06: Camp expenses shall be linked to the relevant budget and expenditure records.

FR-CAMP-07: The system shall record completion, absence, partial participation and approval status.

FR-CAMP-08: The system shall generate camp-wise participant lists, attendance summaries and final reports.

FR-CAMP-09: Camp hours shall be included in cumulative service records without duplicate credit.

3.7 Module M07 — Volunteer Performance and Engagement

FR-PERF-01: The system shall calculate volunteer performance indicators.

FR-PERF-02: Indicators may include:

- Approved service hours.
- Number of attended events.
- Attendance consistency.
- Participation in outreach activities.
- Training completion.
- Quiz scores.
- Verified leadership responsibilities.
- Timely submission of assigned reports.

FR-PERF-03: Authorized administrators shall configure the weighting and eligibility rules.

FR-PERF-04: The system shall provide individual and unit-wise performance dashboards.

FR-PERF-05: The system shall identify volunteers who may need encouragement or follow-up.

FR-PERF-06: The system shall support Outstanding, Highly Active, Consistent, Needs Encouragement and Inactive categories, with configurable criteria.

FR-PERF-07: Programme Officers shall be able to add documented remarks and recommendations.

FR-PERF-08: Performance categories shall not automatically determine certificate eligibility.

FR-PERF-09: The system shall support event participation awards and verified volunteer recognition records.

3.8 Module M08 — Online Quiz and Training

FR-QUIZ-01: Authorized users shall be able to create quizzes and training activities.

FR-QUIZ-02: Supported question types shall include MCQ and True/False in the initial release.

FR-QUIZ-03: Quiz settings shall include start/end date, duration, marks, attempts and passing criteria.

FR-QUIZ-04: The system shall support question banks and reusable questions.

FR-QUIZ-05: Quiz results shall be calculated automatically where applicable.

FR-QUIZ-06: The system shall store attempt history and final scores.

FR-QUIZ-07: The system shall provide unit-wise and individual performance reports.

FR-QUIZ-08: Leaderboards shall support quiz score, participation and other approved metrics.

FR-QUIZ-09: Quiz scores and NSS service hours shall remain separate records.

FR-QUIZ-10: The system shall prevent unauthorized access to answer keys and unpublished questions.

3.9 Module M09 — Certificate Generation, Approval and Verification

FR-CERT-01: The system shall maintain configurable certificate eligibility rules.

FR-CERT-02: Eligibility checks shall include prescribed service hours, required camp participation, service period and other applicable criteria.

FR-CERT-03: The system shall generate an eligibility checklist for each volunteer.

FR-CERT-04: Programme Officers shall be able to review and recommend eligible cases.

FR-CERT-05: The NSS Coordinator and designated approving authority shall process approvals according to the university's approved workflow.

FR-CERT-06: Final certificate issuance shall be restricted to the authorized approving authority or explicitly delegated role.

FR-CERT-07: The system shall generate certificates using an approved template.

FR-CERT-08: Each certificate shall have a unique Certificate ID and verification reference.

FR-CERT-09: Certificates shall include approved student details, certificate type, issue date, authorized signatory details and other required fields.

FR-CERT-10: The system shall maintain Draft, Pending Approval, Approved, Issued, Rejected, Revoked and Superseded statuses.

FR-CERT-11: The system shall support authorized PDF download.

FR-CERT-12: The system shall provide a secure certificate verification interface accessible to authorized users. Any unauthenticated verification endpoint, if required, shall expose only the minimum necessary information and shall be approved separately.

FR-CERT-13: Revocation, correction or reissue shall require a reason, authorized approval and a complete audit trail.

FR-CERT-14: Certificate issuance shall not be treated as a government-issued certificate unless the issuing authority and process have been formally authorized.

3.10 Module M10 — Finance and Budget Management

FR-FIN-01: The system shall maintain an event-wise budget and expenditure record.

FR-FIN-02: Financial records shall include:

- Proposed Budget.
- Requested Amount.
- Sanctioned Amount.
- Released Amount.

- Actual Expenditure.
- Unspent/Available Balance.
- Settlement Status.

FR-FIN-03: Each budget request shall include event, unit, purpose, expense category, amount, requested-by user and supporting documents.

FR-FIN-04: The system shall support configurable approval workflows.

FR-FIN-05: Financial approvals shall follow the university's authorized delegation of powers.

FR-FIN-06: The system shall record sanction reference number, approval date, approving authority and sanction document.

FR-FIN-07: The system shall record release/disbursement details and payment references where applicable.

FR-FIN-08: Actual expenditure entries shall include date, amount, expense category, description, bill/voucher number and supporting documents.

FR-FIN-09: The system shall support bill and voucher upload, review and verification.

FR-FIN-10: Financial records shall support Pending, Submitted, Under Review, Approved, Rejected, Partially Settled and Settled statuses as appropriate.

FR-FIN-11: The system shall calculate balances separately:

- Sanctioned Amount minus Released Amount.
- Released Amount minus Eligible/Recorded Expenditure.
- Total sanctioned amount minus total recorded expenditure, where relevant.

FR-FIN-12: The system shall generate event-wise, unit-wise, category-wise and financial-year-wise reports.

FR-FIN-13: The system shall support budget-versus-actual comparisons.

FR-FIN-14: The system shall prevent unauthorized deletion or alteration of approved financial entries.

FR-FIN-15: Corrections shall require authorized adjustment entries, reasons and audit logs.

FR-FIN-16: The system shall support export of financial summaries to PDF and Excel.

FR-FIN-17: The system shall maintain a clear distinction between an internal financial record and an official accounting or government payment record.

3.11 Module M11 — Inventory and Asset Management

FR-INV-01: The system shall maintain an item master with unique Item ID.

FR-INV-02: Item fields shall include name, category, unit of measurement, quantity, stock threshold, asset status and relevant purchase details.

FR-INV-03: The system shall record stock receipts, purchases, issues, returns, transfers, adjustments and disposals.

FR-INV-04: Every stock transaction shall include date, quantity, responsible user, source/destination and remarks.

FR-INV-05: Inventory may be linked to an event, unit, camp or authorized custodian.

FR-INV-06: The system shall maintain stock balance after every approved transaction.

FR-INV-07: The system shall issue low-stock alerts when the configured threshold is reached.

FR-INV-08: The system shall record damaged, lost or obsolete items.

FR-INV-09: Purchase records shall link to the relevant bill, voucher and expenditure entry.

FR-INV-10: The system shall generate Stock Register, Asset Register, Issue Register and Purchase Reports.

FR-INV-11: High-value or individually identifiable assets shall support asset tag numbers and custodian assignment.

3.12 Module M12 — News, Gallery and Social Media Records

FR-MEDIA-01: Authorized users shall be able to create internal news and announcements.

FR-MEDIA-02: The system shall store event photographs, videos and supporting documents.

FR-MEDIA-03: Media records shall be linked to events and units.

FR-MEDIA-04: The system shall support approval status and content version history.

FR-MEDIA-05: Social media records shall include platform, post URL, publication date and responsible user.

FR-MEDIA-06: Engagement data may include likes, comments, shares, reach and clicks where available.

FR-MEDIA-07: Engagement data shall identify whether values were manually entered or obtained through an authorized integration.

FR-MEDIA-08: Automatic Instagram or other social platform integration shall require approved API access and permissions.

FR-MEDIA-09: The module shall remain internal and shall not create a public-facing website in the initial release.

3.13 Module M13 — Notifications and Communication

FR-NOT-01: The system shall provide in-portal notifications.

FR-NOT-02: Notifications shall support event registration, upcoming activities, attendance verification, pending tasks and certificate status.

FR-NOT-03: Authorized users shall be able to issue announcements to selected units or volunteers.

FR-NOT-04: Email or other messaging integrations shall be configurable and permission-controlled.

FR-NOT-05: The system shall maintain notification status and delivery information where supported.

FR-NOT-06: The system shall generate alerts for configurable inactivity or incomplete service-hour targets.

3.14 Module M14 — Reports and Analytics

FR-REP-01: The system shall provide a consolidated NSS Coordinator dashboard.

FR-REP-02: Reports shall support filtering by academic session, date range, unit, school, event category and status.

FR-REP-03: The system shall provide:

- Volunteer Master Report.
- Unit-wise Enrollment Report.
- Event and Activity Report.
- Attendance Report.
- Individual Service-Hour Statement.
- Annual and Cumulative Hours Report.
- Special Camp Report.
- Certificate Eligibility and Issuance Report.
- Volunteer Performance Report.
- Inactive Volunteer Report.
- Budget and Expenditure Report.
- Bill/Voucher Status Report.
- Inventory and Asset Report.
- News and Media Documentation Report.
- Annual NSS Activity Summary.

FR-REP-04: Reports shall be exportable to PDF, Excel or CSV according to user permissions.

FR-REP-05: Annual reports shall consolidate approved data and provide downloadable supporting tables.

FR-REP-06: All reports shall distinguish draft, pending and approved records where relevant.

FR-REP-07: Dashboard statistics shall be calculated from the underlying records and shall not rely on manually maintained totals.

3.15 Module M15 — Audit Trail and Data Governance

FR-AUD-01: The system shall log important create, update, approve, reject, issue and revoke actions.

FR-AUD-02: Audit records shall include user, action, record reference and timestamp.

FR-AUD-03: Financial and certificate actions shall include approval and reason details.

FR-AUD-04: Approved attendance and service-hour corrections shall preserve the original values and adjustment history.

FR-AUD-05: Audit records shall not be editable by ordinary users.

FR-AUD-06: The system shall support archival of previous academic sessions without losing historical information.

FR-AUD-07: Data retention, archival and deletion shall follow university policy and applicable law.

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4. Core Business Rules

BR-01: Volunteer Capacity

The initial system shall support six units with a default capacity of 100 volunteers per unit. The capacity shall be configurable by an authorized administrator.

BR-02: Service-Hour Rules

The system shall support configurable annual, cumulative and activity-specific service-hour requirements. The 120-hour and 240-hour thresholds shall be configured only after verification of the applicable NSS rules.

BR-03: Approved Hours

Only approved service-hour ledger entries shall count towards the official completion total.

BR-04: No Duplicate Credit

A volunteer shall not receive duplicate credit for the same attendance record or service contribution.

BR-05: Special Camp

The prescribed special-camp duration and eligibility criteria shall be configurable. Completion shall be recorded separately from regular activity hours.

BR-06: Certificate Eligibility

Meeting a numerical hour target shall not, by itself, guarantee certificate eligibility. All applicable requirements and approvals must be satisfied.

BR-07: Certificate Authorization

Only an authorized role shall issue, revoke or reissue certificates.

BR-08: Financial Controls

Requested, sanctioned, released and spent amounts shall be stored separately. A budget request shall not be treated as an approved sanction.

BR-09: Financial Balances

Balances shall be calculated from the relevant approved financial records. The system shall not silently overwrite historical amounts.

BR-10: Inventory Controls

Inventory balances shall be derived from approved stock transactions rather than independent manual totals.

BR-11: Performance Indicators

Volunteer performance scores shall use transparent, configurable criteria. Performance ranking shall not automatically determine certification.

BR-12: Historical Records

Changes to unit allocation, activity categories or annual settings shall not alter historical approved records without an authorized adjustment process.

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5. Proposed Database Design

The database shall use a relational structure with primary keys, foreign keys, unique constraints and transaction controls.

5.1 Core Tables

Table	Purpose
users	Authentication and account information
roles	Role definitions
permissions	Fine-grained permissions
user_roles	User-to-role mapping
volunteers	Volunteer profile and NSS identity
nss_units	Unit master and capacity
academic_sessions	Academic and reporting periods
volunteer_unit_history	Unit allocation history
events	Activity and event master
event_units	Units participating in an event
event_registrations	Volunteer registration records

attendance	Event-wise attendance
service_hour_ledger	Approved service-hour entries and adjustments
special_camps	Special-camp master
camp_days	Day-wise camp schedule
camp_participation	Volunteer camp participation and attendance
quizzes	Quiz master
quiz_questions	Question bank
quiz_options	Question answer options
quiz_attempts	Volunteer attempts and scores
certificates	Certificate details and status
certificate_approvals	Approval workflow and decision history
budget_requests	Event budget requests
budget_sanctions	Approved sanctions and references
fund_releases	Release/disbursement records
expenditures	Actual expenditure entries
bills_vouchers	Supporting bills, vouchers and verification
inventory_items	Item master
stock_transactions	Purchases, issues, returns and adjustments
assets	Individually identifiable assets
asset_custody	Asset custodian and assignment history
media_records	Photos, videos and documents
news_records	Internal announcements and news
social_media_records	Post links and engagement metrics
notifications	Notification records
audit_logs	Security and operational audit trail
attachments	Secure file metadata and record references

5.2 Data Integrity Requirements

- Every entity shall have a stable unique identifier.
- Foreign keys shall preserve relationships between records.
- Student identifiers shall be unique according to university-approved rules.
- Duplicate attendance and service-hour entries shall be prevented.
- Financial and inventory transactions shall use database transactions where needed.
- Approved financial entries shall not be hard-deleted.
- Attachments shall reference authorized records and be subject to access checks.
- Academic-session and historical records shall remain traceable.

The final database schema and Entity Relationship Diagram (ERD) shall be submitted by the developer before production implementation.

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6. Main User Workflows

6.1 Volunteer Registration

1. Volunteer submits the registration form.
2. System validates required fields and checks for duplicates.
3. Application enters Pending status.
4. Authorized Programme Officer or Coordinator reviews the application.
5. Application is approved, rejected or waitlisted.
6. Approved volunteer receives a unique Volunteer ID and unit allocation.
7. The volunteer can access the personal dashboard.

6.2 Event and Attendance Workflow

1. Programme Officer creates an event.
2. Event is reviewed and approved according to the configured policy.
3. Event becomes available for registration.
4. Volunteers register or are assigned to the event.
5. Attendance is recorded during or after the event.
6. Programme Officer verifies attendance and proposed hours.
7. Approved hours are entered into the service-hour ledger.
8. Dashboard statistics and reports are updated.
9. Supporting documents remain linked to the event.

6.3 Special Camp Workflow

1. Authorized officer creates the camp.
2. Participating units and volunteers are recorded.
3. Daily schedules and activities are entered.
4. Daily attendance is recorded and verified.
5. Daily reports, photographs and expenses are uploaded.
6. Camp completion is reviewed.
7. Approved camp participation is reflected in the volunteer record.
8. The final camp report is generated.

6.4 Certificate Workflow

1. System calculates preliminary eligibility.
2. Programme Officer reviews the volunteer's service history.
3. Missing hours or other requirements are identified.
4. Eligible cases are recommended.
5. NSS Coordinator reviews the recommendation.
6. The designated approving authority approves or rejects the case.
7. An authorized user generates and issues the certificate.
8. Certificate ID, issue date, approval reference and PDF are stored.
9. The certificate can be verified using its unique reference.
10. Any subsequent correction or revocation is recorded in the audit trail.

6.5 Financial Workflow

1. Programme Officer submits an event budget request.
2. Authorized reviewer examines the request.
3. Competent authority approves or rejects the sanction.
4. Approved sanction reference and amount are recorded.
5. Release/disbursement details are entered.
6. Actual bills and vouchers are uploaded against the event.
7. Finance/Accounts verifies the documents.
8. Expenditure is recorded and categorized.
9. Outstanding documents and balances are calculated.
10. The event is settled after completion of the required review.

6.6 Inventory Workflow

1. Item master is created.
2. Purchase or receipt is recorded.
3. Supporting bill is linked where applicable.
4. Stock quantity is updated.
5. Items are issued to a unit, custodian or event.
6. Returnable items are returned and verified.
7. Damaged, lost or disposed items are documented.
8. Stock and asset reports are generated.

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7. Dashboard Requirements

7.1 NSS Coordinator Dashboard

- Total registered volunteers.
- Unit-wise enrollment and capacity.
- Active and inactive volunteers.
- Events completed and upcoming.
- Total approved service hours.
- Volunteers approaching or completing hour targets.
- Special-camp completion.
- Pending attendance approvals.
- Pending certificate cases.
- Top-performing volunteers.
- Unit-wise performance comparison.
- Pending reports and documentation.
- Total sanctioned budget, releases and expenditure.
- Pending bills and financial settlements.
- Inventory stock alerts.

7.2 Programme Officer Dashboard

- Unit-specific volunteer list.
- Upcoming and completed activities.
- Attendance awaiting verification.
- Volunteer service-hour progress.

- Volunteers needing follow-up.
- Pending event reports.
- Special-camp participation.
- Certificate recommendations.
- Unit-level financial and inventory information, subject to permissions.

7.3 Volunteer Dashboard

- Personal profile and NSS ID.
- Assigned unit and Programme Officer.
- Upcoming activities.
- Event registration history.
- Attendance history.
- Approved, pending and remaining service hours.
- Special-camp participation.
- Quiz scores and leaderboard position.
- Certificate eligibility checklist.
- Issued certificates and downloads.
- Notifications and announcements.

7.4 Finance Dashboard

- Budget requests awaiting review.
- Total sanctioned amounts.
- Total released amounts.
- Recorded expenditure.
- Available balances.
- Bills awaiting verification.
- Pending settlements.
- Event-wise and unit-wise expenditure.
- Financial-year summaries.

7.5 Inventory Dashboard

- Total items and assets.
- Available stock.
- Issued and returnable items.

- Low-stock alerts.
- Damaged/lost items.
- Recent purchases and stock movements.
- Item-wise and event-wise reports.

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8. Non-Functional Requirements

8.1 Security

- Enforce role-based access control on the frontend and backend.
- Use HTTPS in production.
- Store passwords using a suitable secure password-hashing algorithm.
- Protect against SQL injection, cross-site scripting, cross-site request forgery and unauthorized file access.
- Validate uploaded files by type, size and content.
- Restrict access to financial documents and certificates.
- Implement secure session handling, rate limiting and appropriate login protections.
- Keep secrets and credentials outside source code.

8.2 Performance

- Target a response time of under three seconds for ordinary dashboard and search operations under the agreed test load.
- Use pagination for large lists.
- Optimize database queries and indexes.
- Process large reports and bulk imports without blocking routine operations.

8.3 Availability and Reliability

- Provide scheduled database backups.
- Define and test restoration procedures.
- Use appropriate error handling and transaction rollback.
- Prevent partial financial and inventory updates.
- Document deployment, recovery and maintenance procedures.

8.4 Scalability

- Support the initial 600 volunteers.
- Permit expansion to additional units and larger volunteer populations.
- Avoid hard-coded unit numbers, capacities and service-hour thresholds.

8.5 Usability and Accessibility

- Responsive interface for mobile, tablet and desktop.
- Clear labels and validation messages.
- Search, filtering and pagination.
- Accessible forms and sufficient contrast.
- Consistent date, currency and status formats.

8.6 Auditability

- Maintain immutable or appropriately protected audit records.
- Record important changes to attendance, hours, certificates, budgets and inventory.
- Preserve approval and adjustment history.

8.7 Data Privacy

- Collect only necessary personal information.
- Restrict personal and sensitive information to authorized users.
- Define retention and archival rules.
- Use approved institutional storage for confidential documents.
- Avoid exposing student-level records in public or unauthenticated interfaces.

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9. Proposed Technical Architecture

The developer may propose an equivalent stack, subject to approval by the university's IT Cell.

Frontend

- React or Next.js.
- Responsive UI using Tailwind CSS or an equivalent component system.
- Charting library for dashboards.

Backend

- Node.js with Express or NestJS, or another approved backend framework.
- REST API with documented request and response schemas.
- Centralized business rules and permission checks.

Database

- PostgreSQL recommended for relational data integrity and financial workflows.

File Storage

- Secure, access-controlled storage for photos, reports, bills, vouchers and certificates.

- Files should not be publicly accessible by default.

Reporting

- PDF generation.
- Excel/CSV exports.
- Unit-wise, event-wise and annual summaries.

Deployment

- University-approved server or approved cloud infrastructure.
- Separate development, testing and production environments.
- Version-controlled source code and documented deployment configuration.

Architecture Principles

- The frontend shall not directly modify database records.
- All sensitive operations shall pass through authenticated backend APIs.
- Business rules shall be implemented centrally.
- Database changes shall use version-controlled migrations.
- Configuration values shall be manageable without changing application code where appropriate.

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10. Integration Requirements

Phase 1

- Email notifications, if an approved mail service is available.
- PDF and Excel generation.
- CSV/Excel data import.
- QR-based attendance, subject to implementation approval.

Optional Later Integrations

- University ERP/student information system.
- Official Instagram/Meta API.
- SMS or approved messaging gateway.
- Institutional identity or single sign-on.
- External certificate-verification interface, if authorized.

No integration shall assume access to private university or government systems without credentials, authorization and a defined interface.

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11. Testing and Acceptance Criteria

The system shall be considered ready for acceptance only after the following tests pass.

Volunteer Management

- Six units can be created with configurable capacities.
- Registration, approval and allocation work correctly.
- Duplicate student registrations are rejected or flagged.
- Historical volunteer records remain accessible.

Activity and Attendance

- An event can be created, approved, published and completed.
- Multiple units can participate in the same event.
- Attendance can be recorded and verified.
- Approved hours are calculated correctly.
- Duplicate service-hour credits are prevented.
- Corrections preserve the original value and audit history.

Special Camp

- Day-wise attendance and activities can be recorded.
- Camp reports and supporting documents can be generated.
- Camp completion status is correctly reflected in volunteer records.

Performance and Quiz

- Performance metrics are derived from approved records.
- Quiz scores are calculated correctly.
- Quiz results do not automatically increase service hours without an approved rule.

Certificate Management

- Eligibility is calculated using configured rules.
- Unauthorized users cannot issue certificates.
- Approved certificates have unique identifiers.
- Verification returns the correct certificate status.
- Revoked or superseded certificates cannot be presented as currently valid.

Finance and Inventory

- Budget requests and sanctions are distinguishable.
- Released amounts and expenditure are calculated correctly.

- Bills and vouchers can be linked to relevant transactions.
- Approved financial records cannot be silently deleted or modified.
- Stock receipts, issues and returns produce correct balances.
- Inventory purchases can be linked to expenditure records.

Reporting and Security

- Dashboard figures match underlying records.
- PDF/Excel exports respect user permissions.
- Unauthorized access to protected records is blocked.
- Audit logs capture critical actions.
- Backup restoration is tested.
- No critical or high-severity security defects remain unresolved at production release.

The university shall approve a detailed test plan and user acceptance testing (UAT) checklist before final sign-off.

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12. Development Deliverables

The developer shall deliver:

1. Approved SRS and requirement traceability matrix.
2. UI/UX wireframes and final interface designs.
3. Database schema and Entity Relationship Diagram.
4. Backend API specification.
5. Working frontend and backend source code.
6. Role-specific dashboards and access permissions.
7. Database migration scripts and initial configuration.
8. Attendance, service-hour and special-camp modules.
9. Certificate generation and approval workflow.
10. Finance, bills, vouchers and inventory modules.
11. Reports, analytics and export functions.
12. Test cases and UAT results.
13. Security testing report and resolved-defect list.
14. Deployment and configuration documentation.
15. Backup and restoration documentation.

16. Administrator and Programme Officer user manuals.
17. Source-code repository access and university-controlled ownership/administrative access.
18. Handover training and post-deployment support terms.

The university shall retain access to its source code, database, deployment configuration and institutional data, subject to the approved development contract.

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13. Recommended Implementation Phases

Phase 1 — Core Operations

- Authentication and role management.
- Volunteer registration and unit allocation.
- Event management.
- Attendance and service-hour ledger.
- Special-camp management.
- Core Coordinator, Programme Officer and Volunteer dashboards.
- Basic reports and audit logs.

Phase 2 — Certification and Administration

- Performance analytics.
- Online quizzes and leaderboard.
- Certificate eligibility, approval, generation and verification.
- Finance, budgets, sanctions, bills and vouchers.
- Inventory and asset management.
- Advanced reports and annual summaries.

Phase 3 — Engagement and Optimization

- Notifications and reminders.
- Internal news and media management.
- Social media records and authorized integrations.
- Advanced analytics.
- Optional ERP or identity integration.
- Performance optimization and additional automation.

Each phase shall have defined milestones, test cases and acceptance sign-off.

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14. Assumptions and Items Requiring Institutional Approval

The following items must be confirmed by the university or the authorized NSS authority before final production configuration:

1. Applicable official NSS rules for annual and cumulative service hours.
2. Required duration and completion criteria for the special camp.
3. Certificate format, issuing authority, authorized signatory and verification requirements.
4. Delegation of financial sanction and approval powers.
5. Approved expense categories and required financial documents.
6. Data retention, privacy and archival policies.
7. University-approved deployment environment and backup policy.
8. Whether any external integrations are permitted.
9. Academic-session rollover and volunteer continuation rules.
10. Final performance scoring criteria and recognition policy.

These values shall be configurable wherever technically appropriate.

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15. Final Project Outcome

The completed NSS Management Portal shall provide a centralized, secure and auditable platform for managing the entire NSS volunteer lifecycle at Gautam Buddha University.

It shall integrate volunteer records, activities, attendance, verified service hours, special camps, performance, online quizzes, certificates, event budgets, expenditure, bills, vouchers, inventory and institutional reports.

The system shall reduce manual registers and spreadsheets, improve accountability, enable evidence-based volunteer recognition, simplify certificate processing and provide reliable information for administrative review and annual reporting.

End of SRS Document — Version 1.0